DATASCI 406 FA 26, Quiz 1.4

Name and Uniqname:

Question 1 (1 point)

You simulate $B = 999$ datasets under H_0 and find that **every** simulated value of the test statistic is smaller than the observed t_{obs}. What do you report? **Circle one.**

1. $\hat{p} = 0$
2. $\hat{p} = 1/1000$
3. $\hat{p} = 1/999$
4. $\hat{p} < 0.05$; the simulation cannot pin the value down any further.

Question 2 (1 point)

$X_1, \ldots, X_{25}$ are independent draws from $N(\mu, 1)$; the statistic T is the **sample median**; $H_0 : \mu = 0$ against $H_1 : \mu = 0.35$; and the rejection region was fixed in advance as $T > 0.3$.

```
set.seed(406)
null <- replicate(20000, median(rnorm(25, 0.00, 1)))
alt  <- replicate(20000, median(rnorm(25, 0.35, 1)))
round(c(A = mean(alt),      B = mean(null > 0.3),
        C = mean(alt < 0.3), D = mean(alt > 0.3)), 3)
```

```
##     A     B     C     D
## 0.352 0.116 0.420 0.580
```

Which printed value is the **size** of this test, and which is its **power** against $\mu = 0.35$? Write one label in each blank.

size = _____ power = _____

Question 3 (3 points)

$X_1, \ldots, X_{40}$ are independent Exp(rate $= \theta$) with θ unknown. The test statistic is the 25% trimmed mean `mean(x, trim = 0.25)`, which has no convenient sampling distribution. The observed data give $t_{\text{obs}} = 1.39$.

Suppose we have implemented a **two-sided level-0.05 test** of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$, for one particular θ_0 using a Monte Carlo hypothesis test. The procedure is implemented as a function `test(t_obs, theta_0)` that returns `TRUE` if the hypothesis is rejected and `FALSE` if the hypothesis is not rejected.

Using this function, describe how you would produce a **95% confidence interval for** θ. Your answer must say both (i) what you would compute, in enough detail that someone could code it from your description, and (ii) why what you get is a 95% interval. You may suppose that θ is known to fall in some range $[a, b]$.